Heuristic Search in Business Intelligence (BI)

In Business Intelligence, companies often face huge amounts of data and many possible decisions.

A heuristic search helps managers and analysts make faster, practical decisions by using rules-of-thumb instead of checking every possible option.

It gives a good enough solution quickly, even if it's not always the absolute best.

Simple Example

Scenario:
A retail store wants to decide which products to promote during a holiday sale.

Exact/Exhaustive Search (hard way):
Analyze every product's sales history, customer preferences, seasonal patterns, competitor prices, etc., for thousands of products. This takes too much time.

Heuristic Search (smart way):
Use a simple rule like:

"Promote products that had at least 20% sales growth last holiday season and profit margin above 30%."

Instead of analyzing all products in detail, the heuristic quickly narrows down to a smaller set of promising products.

Another BI Example: Customer Targeting

A company wants to send discount coupons to customers.

Heuristic Rule (Search shortcut):
"Select customers who purchased at least 3 times in the last 2 months and spent more than ■5,000."

This doesn't guarantee it finds all the best customers, but it's a fast, reasonable way to identify a valuable group without deep analysis.

Summary

Heuristic Search in BI = using rules-of-thumb (heuristics) to quickly search and filter through massive data to support decisions.

It saves time and resources by avoiding exhaustive analysis.

Example: Retailers choosing products to promote or customers to target using simple shortcut rules instead of analyzing every possibility.